Dynamic Exponent-Window Accumulation with Activation Top-2 Selection for Efficient Low-Bit BFP LLM Inference

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***Abstract*—Block Floating Point (BFP) enables low-cost fixed-point multiply–accumulate within each block, but inter-block accumulation still requires an expensive floating-point accumulator (FP-Acc) when shared exponents differ. In synthesized low-bit BFP processing elements, the FP-Acc can consume up to 75% of power and 55% of area. Activation outliers further widen the exponent dynamic range and amplify FP-Acc activity. We propose Dynamic Exponent-Window Accumulation (DEWA), which confines most partial sums to a low-cost accumulator with a narrow dynamic range before forwarding exceptions to the FP-Acc. To mitigate outlier-induced window shifting, we combine DEWA with activation-only dual-exponent encoding and a Top-2 outlier cap per block, while keeping weights in single-exponent BFP. On LLaMA2-7B with WikiText-2, the proposed W-BFP4 / Top-2 A-BiE4 configuration preserves accuracy within 0.02 perplexity of a strong Top-2 baseline while substantially reducing FP-Acc utilization in functional simulation.**

***Index Terms*—Block floating point, large language models, quantization, hardware accelerator, outlier handling**

I. Introduction

Low-precision quantization is essential for deploying large language models (LLMs) under tight memory and energy budgets. Block Floating Point (BFP) offers a favorable accuracy–cost tradeoff by assigning a shared exponent to each fixed-size block while storing only low-bit mantissas [1], [2]. Recent BFP accelerators perform intra-block multiply–accumulate (MAC) with fixed-point arithmetic, but inter-block partial sums with mismatched exponents must still be merged in a floating-point accumulator (FP-Acc) [3]–[5].

Two coupled bottlenecks limit the efficiency of BFP-based LLM inference. First, the FP-Acc dominates processing-element (PE) cost: in synthesized low-bit configurations it can account for up to 75% of power and 55% of area [3]–[5]. Second, activation outliers compress the effective mantissa of co-located normal values and widen the exponent spread of partial sums, increasing FP-Acc activity [6]–[9].

This paper develops a hardware-friendly co-design that addresses both bottlenecks jointly. Our contributions are:

- **Dynamic Exponent-Window Accumulation (DEWA):** a low-cost accumulator that retains partial sums within a sliding exponent window and forwards only out-of-window updates to the FP-Acc.
- **Activation-only dual-exponent encoding with Top-2 selection:** weights remain single-exponent BFP while activations use a bounded dual-exponent format that retains at most two outliers per block, reducing storage and MAC control overhead relative to full BiE [6], [7].
- **Evaluation on LLaMA2-7B:** we quantify perplexity, outlier occupancy, and FP-Acc request reduction under a consistent WikiText-2 protocol, showing negligible accuracy loss for the proposed W-BFP4 / Top-2 A-BiE4 design point.

The remainder of this paper reviews BFP deployment challenges (Section II), presents DEWA and Top-2 activation routing (Section III), reports experimental results (Section IV), and concludes (Section V).

II. Background

A. Block Floating Point

Low-precision integer quantization reduces storage and arithmetic cost, but a single tensor-level scaling factor can be dominated by rare large-magnitude values. This issue is particularly severe in LLM activations, where a small number of outliers may compress most nonoutlier values into a limited set of quantization levels and degrade low-bit accuracy [10]–[12].

Block Floating Point (BFP) provides a more fine-grained accuracy–cost tradeoff by partitioning tensors into fixed-size blocks and assigning an independent shared exponent to each block. Each element stores only a sign and a low-bit mantissa, while the exponent overhead is amortized across the block. Intra-block computation requires only fixed-point multiply–accumulate operations, enabling BFP to achieve higher accuracy at a cost comparable to integer quantization [1], [2].

B. Deployment Challenges

Applying BFP to LLM inference acceleration introduces two major bottlenecks.

(1) Inter-block floating-point accumulation. A state-of-the-art BFP processing element (BFP-PE) can perform MAC operations within each block using low-bit fixed-point arithmetic. However, shared exponents differ across blocks, so partial sums cannot be directly accumulated in the integer domain. An FP accumulator (FP-Acc) is therefore required to perform inter-block exponent alignment and floating-point

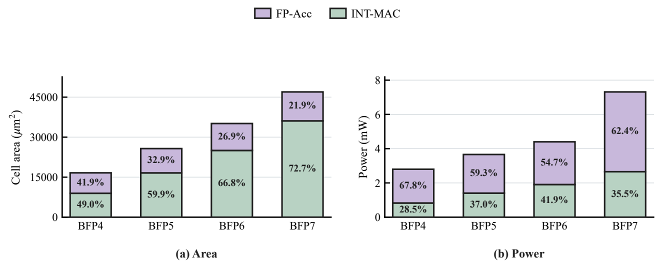

Fig. 1: Component-level power and area breakdown of a synthesized baseline BFP-PE, showing FP-Acc dominance.

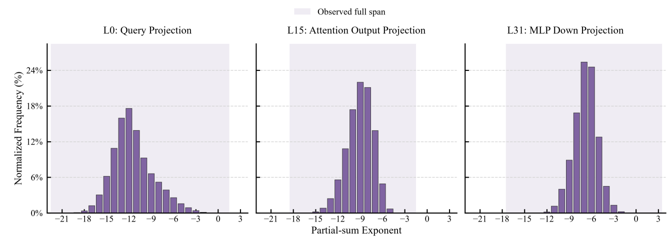

Fig. 2: Representative partial-sum exponent distributions under BFP4 (LLaMA2-7B, G16).

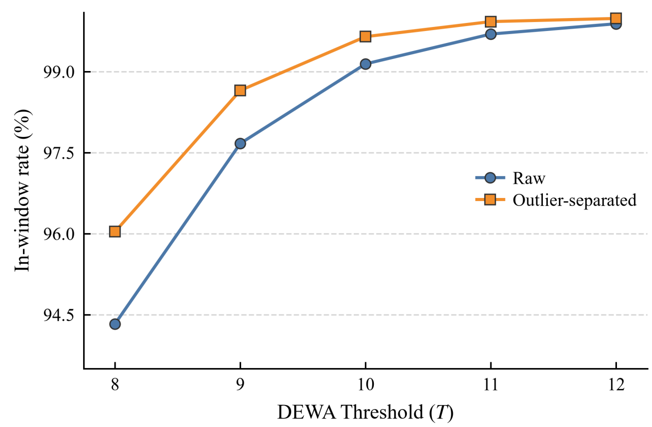

Fig. 3: DEWA in-window rate with and without outlier separation (BFP4, G16, $T = 8$–12).

accumulation. The FP-Acc typically includes alignment, normalization, addition, and fixed-point-to-floating-point conversion logic. In synthesized low-bit BFP-PE configurations, the FP-Acc accounts for up to 75% of PE power and 55% of its area (Fig. 1), making it a primary obstacle to improving the energy and area efficiency of BFP accelerators [3]–[5].

(2) Activation outliers. BFP confines the influence of an outlier to its local block but does not eliminate the outlier itself. When a block contains a large-magnitude outlier, the shared exponent is determined by that value, and nonoutlier values in the same block lose a substantial number of effective mantissa bits [6]–[9]. This loss becomes more severe as the mantissa width decreases.

These two bottlenecks are coupled. Outliers degrade the precision of nonoutlier values during quantization and can also widen the exponent distribution of resulting partial sums, increasing the likelihood that inter-block updates require the FP-Acc. Consequently, effective outlier handling has the potential to improve numerical fidelity while simultaneously reducing FP-Acc activity. This observation motivates the approach developed in this work.

III. Dynamic Exponent-Window Accumulation

A. Partial-Sum Concentration and DEWA

We observed the distribution of partial sums in LLaMA2-7B. Three representative layers under the BFP4 format were selected for illustration. The partial-sum distribution remains highly concentrated, indicating that a low-cost accumulator with a narrow dynamic range can be introduced prior to forwarding data to the FP-Acc.

Specifically, we define an accumulation window

$$[E_{\max} - T_d,\ E_{\max} + T_r] \tag{1}$$

where $E_{\max}$ denotes the exponent of the current maximum partial sum, and T_d and T_r represent the drop and replace thresholds, respectively. When a new partial sum arrives:

- if its exponent falls below the lower bound, it is discarded;
- if it exceeds the upper bound, it directly replaces the value currently held in the accumulator;
- otherwise, it is accumulated using the low-cost accumulator.

This approach substantially reduces FP-Acc utilization and thereby lowers power consumption [3], [4]. We refer to this method as **Dynamic Exponent-Window Accumulation (DEWA)**.

Although the majority of partial sums are concentrated, the overall distribution still exhibits a large dynamic range. We attribute this to the influence of outliers, which cause the dynamic exponent window to shift significantly and thereby induce precision loss. We therefore argue that outlier separation is essential as a complement to DEWA. By separating activation outliers, a substantially higher proportion of partial sums fall within the accumulation window (Fig. 3).

B. Activation Top-2 Selection

In prior work, BiE [6] and Oiso [7] employed a dual-exponent format to separate the exponents of normal values and outliers, prepending a type bit to each element to indicate its category. While this format achieves notable accuracy improvement, it also introduces non-trivial overhead: since the proportion of outliers is inherently low, allocating an additional exponent and type bit to every element—combined with four distinct MAC cases—increases hardware cost. Directly integrating this scheme into DEWA is therefore suboptimal.

Building on this observation, we introduce two improvements:

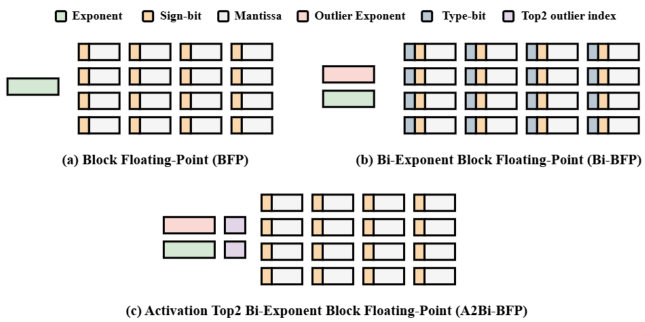

Fig. 4: Data format for W-BFP4 weights and Top-2 A-BiE4 activations (G16).

TABLE I: Outlier occupancy per G16 activation block (W-BFP4/A-BiE4, LLaMA2-7B).

Statistic	Value
Activation outlier rate	0.913% of lanes
Affected blocks	11.40% of blocks
Mean outliers per affected block	1.28
Blocks with exactly one outlier	86.10% of affected
All-block P99 outlier count	2

1) Since the majority of outliers concentrate on specific activation channels, applying dual-exponent encoding to weights yields benefits that do not justify the associated overhead. We therefore adopt single-exponent BFP for weights while retaining dual-exponent encoding for activations.
2) We further examined the number of outliers occurring within each block. Approximately 99.5% of blocks contain at most two outliers. We propose **Activation Top-2 Selection**, wherein at most the two largest outliers are selected per block, further reducing hardware overhead.

Fig. 4 illustrates the resulting W-BFP4 / Top-2 A-BiE4 encoding.

Fig. 5 summarizes the impact of activation-only BiE and Top-2 capping on LLaMA2-7B perplexity, where negligible degradation is observed relative to full BiE.

C. *Hybrid Accumulation Flow*

The canonical hybrid design combines W-BFP4 weights, Top-2 A-BiE4 activations, G16 grouping, asymmetric DEWA thresholds (T_{skip}, T_{replace}), and direct exception routing to the FP-Acc. For each output element within a G16 group, normal and outlier partial sums follow separate accumulation paths before a shared FP-Acc merges exception updates.

IV. Evaluation

A. *Experimental Setup*

We evaluate LLaMA2-7B on the WikiText-2 test split using a non-overlapping 2048-token protocol (166 blocks, 339,802 loss tokens). Fake-quantized weights and activations are dequantized before `F.linear`; attention internals, LayerNorm, and `lm_head` remain in FP16 unless noted. Group size is G16 for the primary design point. BiE-family outlier detection uses $\text{threshold}(X) = \text{mean}(X) + 3\,\text{std}(X)$ over signed tensor values.

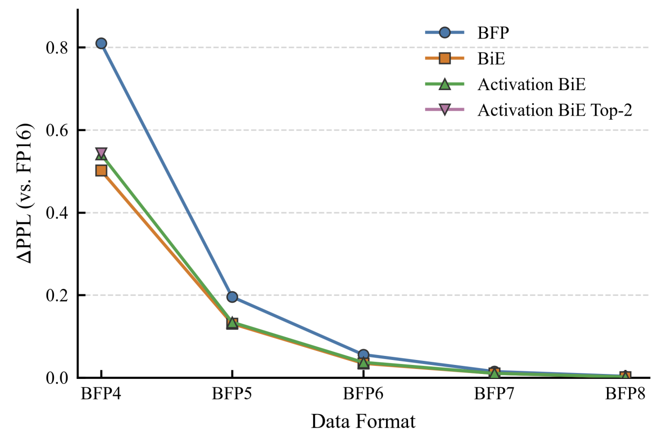

Fig. 5: Perplexity delta versus FP16 baseline for vanilla BFP, BiE, activation-only BiE, and Top-2 activation BiE.

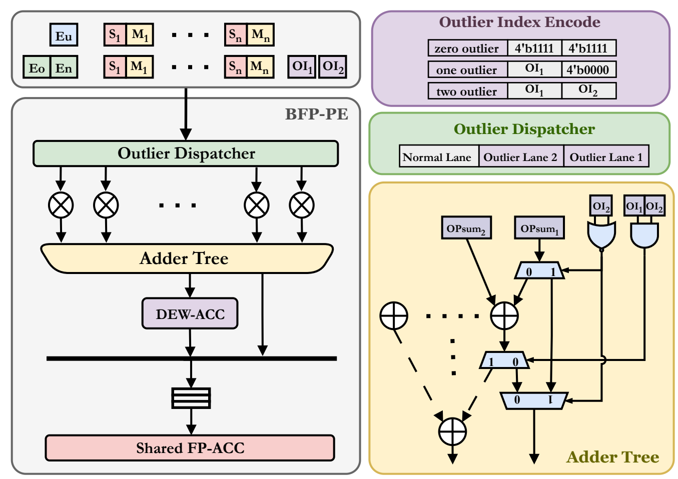

Fig. 6: Proposed Top-2 activation routing with asymmetric DEWA and shared FP-Acc.

B. *Accuracy Results*

Table II compares representative quantization formats against an FP16 reference. Activation-only BiE and Top-2 capping recover most of the accuracy gap of vanilla BFP4 while keeping weights in lower-cost single-exponent BFP.

C. *DEWA Threshold Sweep*

Fig. 7 shows perplexity sensitivity to asymmetric DEWA thresholds under W-BFP4 / Top-2 A-BiE4. A wide operating region near the Top-2 baseline PPL (6.016) indicates that aggressive FP-Acc bypassing is compatible with negligible accuracy loss.

D. *Hardware-Oriented Metrics*

Functional simulation reports DEWA out-of-window (OOW) decisions and optimistic FP-Acc request counts. RTL

TABLE II: WikiText-2 perplexity on LLaMA2-7B (G16, BFP4/BiE4 family).

Configuration	PPL
FP16 baseline	5.472
Vanilla W/A-BFP4	6.283
W/A-BiE4	5.974
W-BFP4 / A-BiE4	6.013
W-BFP4 / Top-2 A-BiE4	6.016

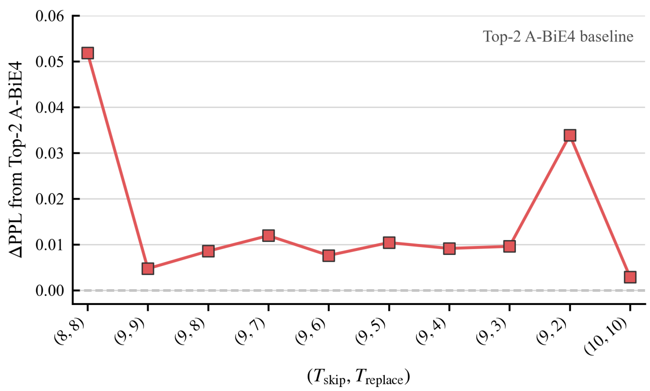

Fig. 7: Perplexity delta versus Top-2 baseline under asymmetric DEWA (T_{skip}, T_{replace}).

synthesis of the baseline BFP-PE confirms FP-Acc dominance (Fig. 1); synthesized area and power for the full Top-2 DEWA design remain future work.

V. Conclusion

BFP accelerators face coupled challenges from expensive inter-block FP accumulation and activation outliers that shift shared exponents and widen partial-sum dynamic range. We presented Dynamic Exponent-Window Accumulation (DEWA) to confine most partial sums to a low-cost accumulator, complemented by activation-only dual-exponent encoding with a Top-2 outlier cap while keeping weights in single-exponent BFP. On LLaMA2-7B, the proposed W-BFP4 / Top-2 A-BiE4 configuration preserves WikiText-2 perplexity within 0.02 of a strong Top-2 baseline while enabling substantial FP-Acc bypass in functional evaluation.

Future work includes RTL implementation and synthesis of the full Top-2 DEWA datapath, cycle-accurate FP-Acc utilization measurement, and extension to additional models and group sizes.

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